M2 ATSI - MASTER E3A
Automatique, Traitement du Signal et des Images
Control, Signal and Image Processing



ATSI Project

Generative DNN for image reconstruction in inverse problems

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I. Introduction

In recent years, deep convolutional networks have proven extremely effective for image generation and restoration, largely due to their ability to learn complex image priors from extensive datasets. However, in our work, we adopt a different perspective, we show that the architecture of convolutional generator networks alone can capture significant low-level image statistics, even without any prior learning. Rather than training on large image collections, we demonstrate that a randomly initialized network can serve as a powerful handcrafted prior for solving classic inverse problems such as denoising, deconvolution, and inpainting, using only a single degraded input image.

In our approach, instead of learning weights from data, the network's parameters are optimized directly to fit the degraded image under a task-specific observation model. This reveals the surprising effectiveness of the network's inherent structure, acting as an implicit prior. Through this method, we achieve competitive results across a variety of image restoration tasks without relying on traditional training. Furthermore, we extend this principle to the analysis of deep network representations : by leveraging an untrained convolutional generator, we enhance techniques such as natural pre-image reconstruction and activation maximization, offering a handcrafted alternative to learned regularizers. This strategy avoids potential biases introduced by learned priors and highlights the inductive biases inherently present in standard convolutional architectures.

II. Procedure

Deep networks are commonly used for image generation by learning generator or decoder networks $x = f_\theta(z)$ that map a random code vector z to an image x, enabling the sampling of realistic images from a learned distribution. In our work, we focus instead on conditional image generation, where the distribution is shaped by a corrupted observation x_0, in order to address inverse problems such as denoising and inpainting.

Our objective is to study the prior inherently encoded by the architecture of a generator network, before any parameter learning occurs. To this end, we interpret the network as a parametrization $x = f_\theta(z)$ of an image, where $x \in \mathbb{R}^{3\times H\times W}$. Here $z \in \mathbb{R}^{C'\times H'\times W'}$ is a random code tensor and θ are the network parameters. The architecture consists of standard operations such as convolution, upsampling, and non-linear activations, and most of our experiments employ a U-Net and Resnet and Autoencoder structure with skip connections, where z and x share the same spatial dimensions. Our default model contains approximately two million parameters.

To validate the effectiveness of this parametrization, we address classic inverse problems, formulating them as energy minimization tasks (instead of maximazing the posterior we minimize this task) :

$$\begin{aligned} x^* &= \arg\max_x p(\hat{x} \mid x)p(x) \\ &= \arg\min_x -\log p(\hat{x} \mid x) - \log p(x) \\ &= \arg\min_x E(x;\hat{x}) + R(x) \end{aligned}$$

where $E\left(x;x_0\right)$ is a data fidelity term based on the observed degraded image x_0, and $R(x)$ is a regularizer typically imposing a natural image prior. Rather than using a traditional hand-crafted regularizer such as Total Variation (TV), we replace $R(x)$ with the implicit prior captured by the untrained neural network itself :

$$\theta^* = \underset{\theta}{\operatorname{argmin}} E\left(f_\theta(z);x_0\right), \quad x^* = f_{\theta^*}(z)$$

Starting from random initialization, we optimize the network parameters θ^* using gradient descent. We keep the input z fixed as a noise tensor with 32 feature maps, matching the spatial resolution of the target image, although in some experiments slight random perturbations to z during optimization improved results.

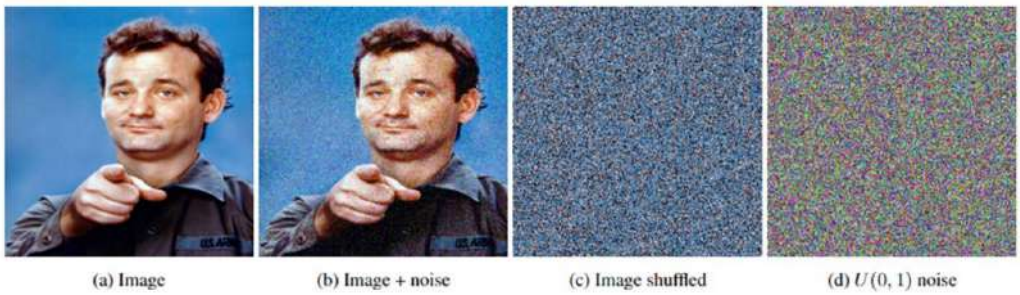
Effectively, the implicit prior defined by this network structure acts as a handcrafted constraint : $R(x) = 0$ for images producible by the generator, and $R(x) = +\infty$ otherwise. Since no pre-training is involved, this "deep image prior" serves a similar handcrafted role as the TV norm, yet captures much richer image statistics. We demonstrate that this simple yet powerful prior leads to strong performance across a variety of image restoration tasks.

A natural question arises : why can a high-capacity network f_θ act as an effective prior at all ? In principle, given its flexibility, one might expect that it should be able to fit any image x, including pure noise, implying that it would impose no meaningful restrictions on the generated images. However, we observe that while the network can indeed eventually fit almost any signal, the architecture strongly influences the optimization dynamics. Specifically, it biases gradient descent towards naturally structured images, resisting convergence to noise-like solutions.

To quantify this effect, we consider a basic reconstruction task : given a target image x_0, we optimize the network parameters θ^* to reproduce it by minimizing :

$$\min_{\theta} \|f_\theta(z) - x_0\|^2$$

where z is a fixed input tensor and $E(x; x_0) = \|x - x_0\|^2$ serves as the data term.



(a) Image (b) Image + noise (c) Image shuffled (d) $U(0, 1)$ noise

FIGURE 1 – Images used in fig. 2 to demonstrate the noise impedance effect.

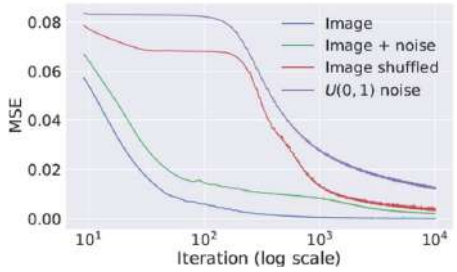


FIGURE 2 – Learning curves for the reconstruction task using : a natural image, the same plus i.i.d. noise, the same randomly scrambled, and white noise. Naturally-looking images result in much faster convergence, whereas noise is rejected.

We evaluate the behavior of this optimization for four types of target images : (1) a clean natural image, (2) the same image corrupted by additive noise, (3) the same image with randomly permuted pixels, and (4) pure white noise. As shown in Figure 2, optimization converges much faster for cases (1) and (2) compared to (3) and (4). This demonstrates that the network parametrization exhibits significant "inertia" or resistance when fitting unstructured noise.

In essence, although the network has enough capacity to fit arbitrary signals, it does so reluctantly for noise-like patterns. The parametrization provides high impedance to noise and low impedance to structured signals. Consequently, for practical applications, we restrict the number of gradient descent iterations when solving (2), effectively projecting the solution onto a subset of images that can be generated by networks whose parameters remain close to their random initialization. This property enables the network to act as a strong implicit prior for natural image restoration tasks.

Optimisation

For optimizing our network step by step we need first come up with the structure of neural network witch is UNet, ResNet or SKIP (Autoencoder with skip connection) the we follow this steps

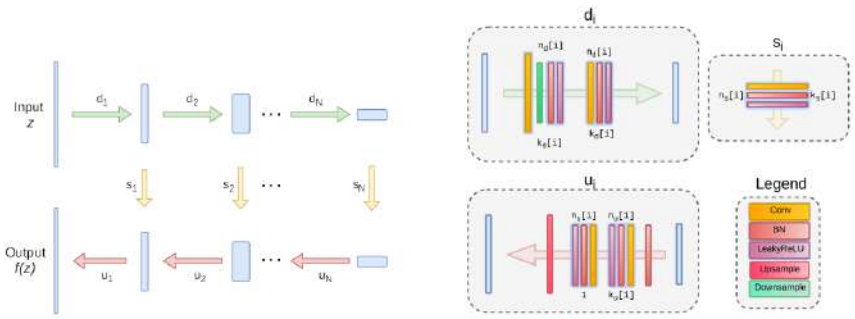


FIGURE 3 – The architecture used in the experiments. We use "decoder-encoder" architecture with skip connection

1. Initialize the input z, for example by filling it with uniform noise $U(-1, 1)$.
2. Solve :

$$\theta^* = \arg\min_{\theta} E\left(f_\theta(z); \hat{x}\right)$$

using a preferred gradient-based method :

$$\theta^{k+1} = \theta^k - \alpha \frac{\partial E\left(f_\theta(z); \hat{x}\right)}{\partial \theta}$$

3. We obtain the solution :

$$x^* = f_{\theta^*}(z)$$

‘

III. Data term

Data term expresses the relation between the clean and the correpted image.

Denoising reconstruction.

we leverage the high noise resistance of our network parametrization to perform image denoising. The objective is to recover a clean image x from a noisy observation x_0. While denoising can sometimes assume a known degradation model (e.g., $x_0 = x + \epsilon$ where ϵ follows a specific distribution).

We maintain the same general optimization framework described earlier, using a randomly-initialized generator network to minimize the discrepancy between the generated image and the noisy input. By solving for the network parameters θ^*, we reconstruct the clean image as $x^* = f_{\theta^*}(z)$.

A key strength of our approach is that it operates without requiring explicit knowledge of the degradation process, making it highly flexible and adaptable. This enables us to apply it directly to restoration tasks where the corruption is complex, poorly understood, or where realistic supervised data is difficult to obtain. We illustrate the robustness of our method through several qualitative examples, including the restoration of images degraded by unknown compression artifacts, as well as cases with convolution.



FIGURE 4 – Noisy Images

Inpainting

For the inpainting task, we consider images x_0 with missing regions, indicated by a binary mask $m \in \{0,1\}^{H \times W}$. The goal is to reconstruct the missing information. To guide this process, we use the data term

$$E\left(x; x_0\right) = \left\|(x - x_0) \odot m\right\|^2,$$

, where $\odot$ denotes element-wise multiplication. Without an additional prior, directly optimizing over the pixel values would leave the missing areas unchanged, highlighting the essential role of a strong prior. As in previous sections, we introduce the prior through our network parametrization, optimizing the data term via the untrained generator.

Deconvolution

For the deconvolution task, we consider images x_0 that have been blurred by an unknown or known kernel. The objective is to recover a sharp version of the original image. To formalize this, we define the data term as

$$E\left(x; x_0\right) = \left\|(k * x) - x_0\right\|^2,$$

represents the convolution of the image x with a blur kernel k. Without an appropriate prior, directly optimizing over pixel values would not adequately restore the fine details lost through blurring. As in previous tasks, we introduce the prior by optimizing this data term through our network parametrization, using an untrained generator to guide the restoration process.

IV. Evaluation and Results

IV.1 Denoising

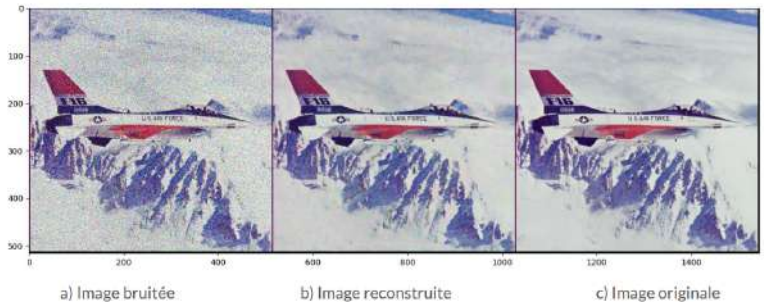


FIGURE 5 – Denoising : SKIP Architecture/ 1500 iterations / noise strength $\sigma = 0.196$

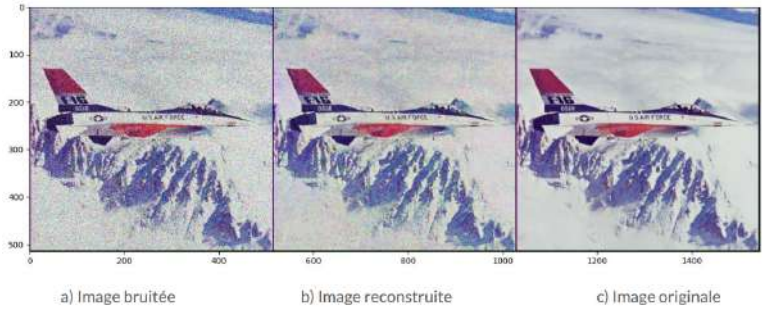


FIGURE 6 – Denoising : SKIP Architecture/ 2500 iterations / noise strength $\sigma = 0.196$

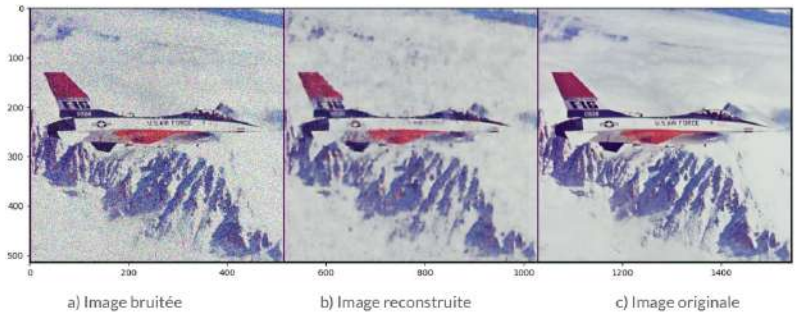


FIGURE 7 – Denoising : ResNet Architecture/ 3000 iterations / noise strength $\sigma = 0.196$

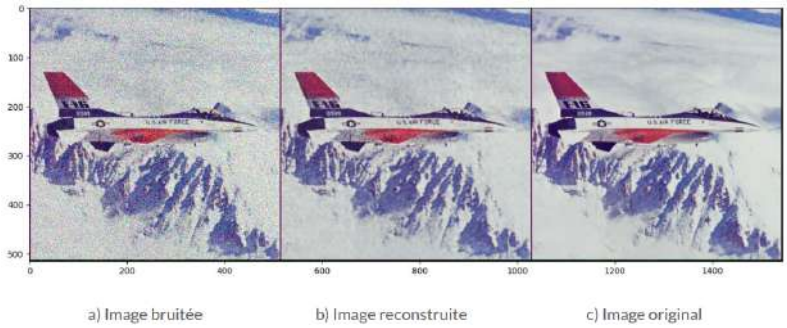


FIGURE 8 – Denoising : UNet Architecture/ 1400 iterations / noise strength $\sigma = 0.196$

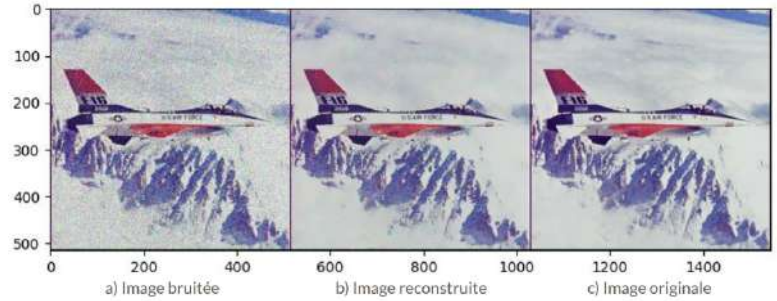


FIGURE 9 – Denoising with NLmeans.

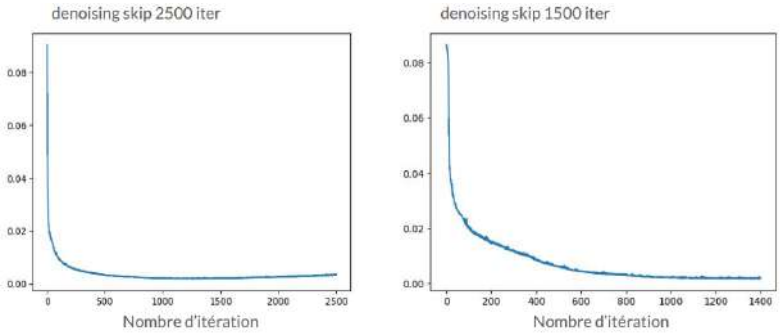


FIGURE 10 – The evolution curve of the parameters and "mse" by the Skip architecture during training.

It can be observed that as the number of iterations increases, the model begins to reconstruct the noise. This is due to the underlying principle of the deep image prior, which is robust to noise. However, after a certain number of iterations, the model starts to reconstruct the noise itself.

Therefore, the training should be stopped earlier, before the model starts reconstructing the noise.

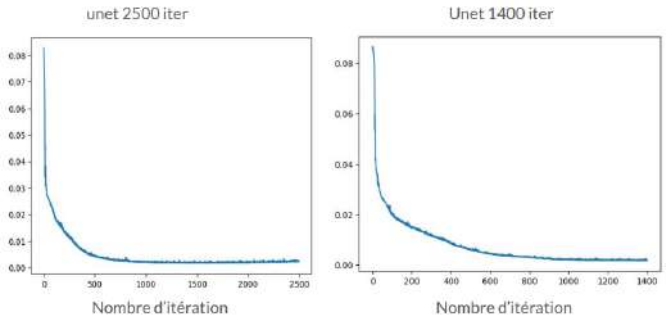


FIGURE 11 – The evolution curve of the parameters and "mse" by the Unet architecture during training.

Comparison

We evaluated the performance of three architectures Skip, ResNet, and U-Net for the denoising task with a noise strength of $\sigma = 0.196$, using different numbers of iterations for each model : 1500 iterations for Skip, 1400 iterations for U-Net, and 3000 iterations for ResNet. The results, shown in the table below, indicate the following.

Denoising	Skip	ResNet	U-Net	NL-means
SSIM	0.73	0.5312	0.723	0.672
MSE	0.00198	0.0043	0.002	0.0043

TABLE 1 – Comparison of Denoising Performance Across Architectures (Skip, ResNet, U-Net)

The **Skip architecture** delivered the best performance, achieving an SSIM of 0.73 and an MSE of 0.00198 after 1500 iterations, showcasing its ability to maintain structural similarity and minimize reconstruction errors. The **U-Net**, trained for 1400 iterations, also performed well, with an SSIM of 0.723 and an MSE of 0.002. Although effective, it slightly lagged behind the Skip architecture, suggesting that the latter might be more efficient in capturing and removing noise in fewer iterations. The **ResNet** model, trained for 3000 iterations, yielded the weakest results, with an SSIM of 0.5312 and an MSE of 0.0043. Finally, the traditional **NL-means** denoising method achieved an SSIM of 0.672 and an MSE of 0.0043, providing moderate performance but still trailing behind the deep learning models, particularly the Skip architecture.

These results emphasize the superior performance of the **Skip architecture**, which outperformed both the traditional NL-means method and the deep learning models (ResNet and U-Net) in terms of both SSIM and MSE, even with fewer iterations. The **Skip network**'s efficiency in denoising, in comparison to ResNet and U-Net, suggests that fewer iterations may be required to achieve better results in this task.

IV.2 Inpainting

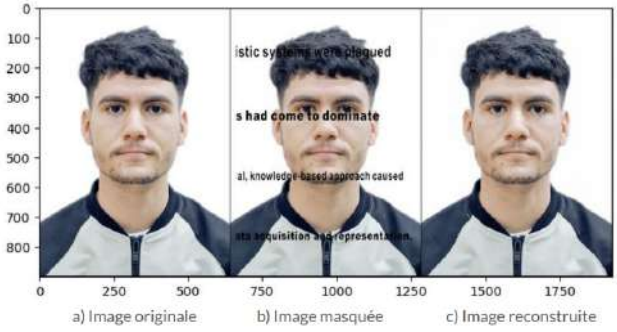


FIGURE 12 – Inpainting : Skip Architecture/ 3000 iterations

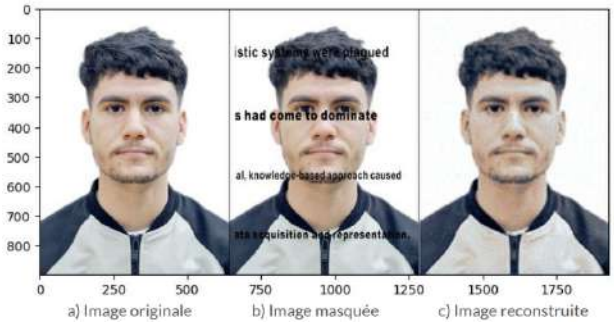


FIGURE 13 – Inpainting : ResNet Architecture/ 3000 iterations

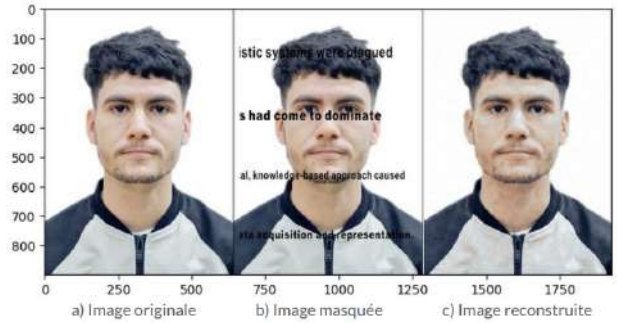


FIGURE 14 – Inpainting : U-Net Architecture/ 3000 iterations

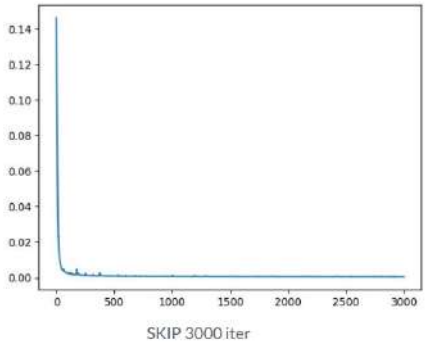


FIGURE 15 – The evolution curve of the parameters and "mse" by the Skip architecture during training.

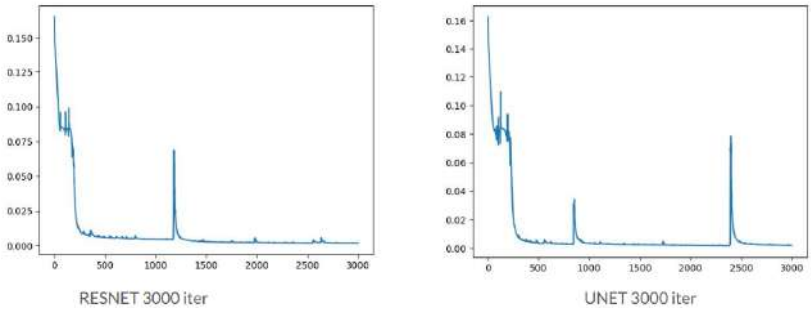


FIGURE 16 – The evolution curve of the parameters and "mse" by the Resnet and Unet architecture during training.

Comparison

We evaluated the performance of three architectures Skip network (an autoencoder with skip connections), ResNet, and U-Net for the task of inpainting a masked handwriting region on a facial image. The evaluation was

based on two metrics : Structural Similarity Index (SSIM) and Mean Squared Error (MSE).

The results are summarized in the following table :

Inpainting	**Skip**	**ResNet**	**U-Net**
SSIM	0.8552	0.7870	0.8235
MSE	0.0013	0.0019	0.0017

TABLE 2 – Comparison of Inpainting Performance Across Architectures (Skip, ResNet, U-Net)

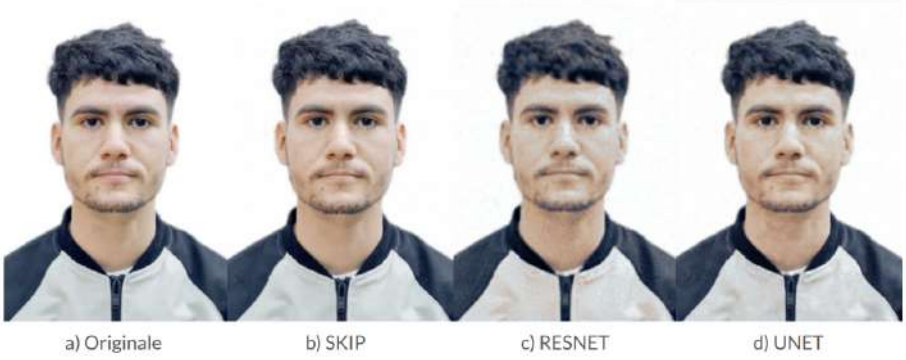


FIGURE 17 – Comparison of the results of the different architectures.

The Skip network achieved the best performance among the three models, with the highest SSIM value (0.8552) and the lowest MSE (0.0013). This suggests that the Skip network was able to better preserve the structural similarity and minimize reconstruction errors compared to ResNet and U-Net. U-Net also performed well, outperforming ResNet but slightly below the Skip network.

ResNet achieved the lowest performance in this experiment, indicating that while it is effective for many tasks, it may not be as well-suited for this specific type of inpainting problem without further adaptation.

Limitation

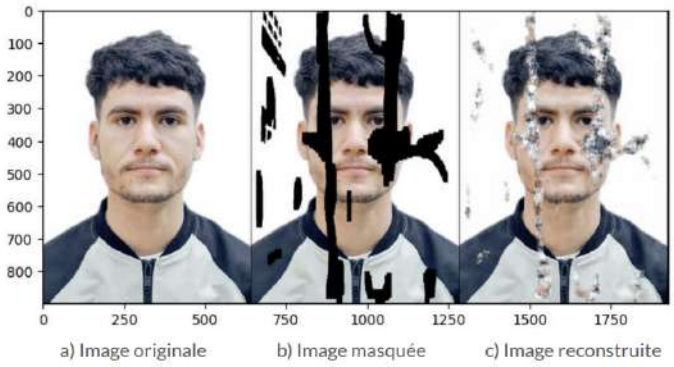


FIGURE 18 – Limitations of DIP : Inpainting Large Missing Facial Regions using 3000 iteration

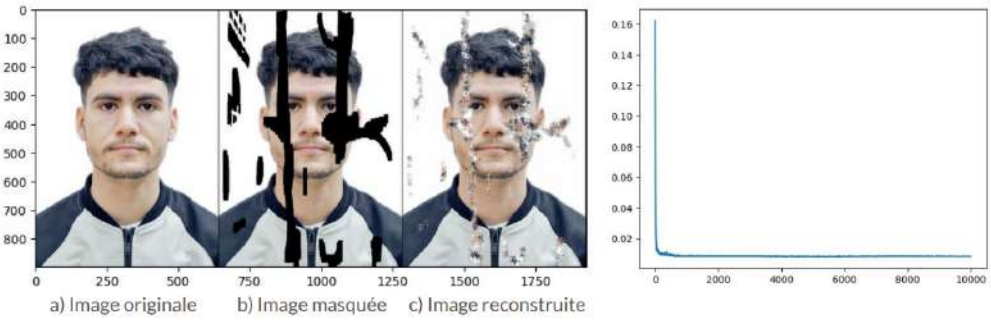


FIGURE 19 – Limitations of DIP : Inpainting Large Missing Facial Regions using 10000 iteration

We applied the Deep Image Prior (DIP) method for image inpainting and obtained very good results when the missing regions were relatively small. However, when using a mask with a large missing area on a facial image, the method failed to properly reconstruct the missing parts. This behavior is expected, as DIP is well-suited for interpolating textures and local image structures but struggles with highly semantic reconstructions, such as facial features (eyes, mouth, etc.). Since DIP does not rely on learned semantic information from datasets but rather exploits the natural image statistics encoded in the network architecture, it cannot hallucinate complex structures that are not present in the surrounding context.

IV.3 Deconvolution

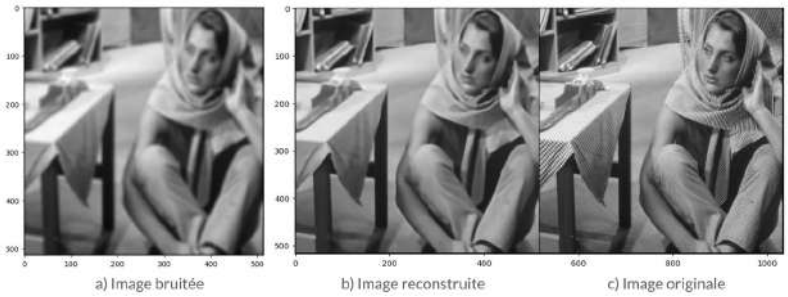


FIGURE 20 – Deconvolution : SKIP Architecture/ 4000 iterations

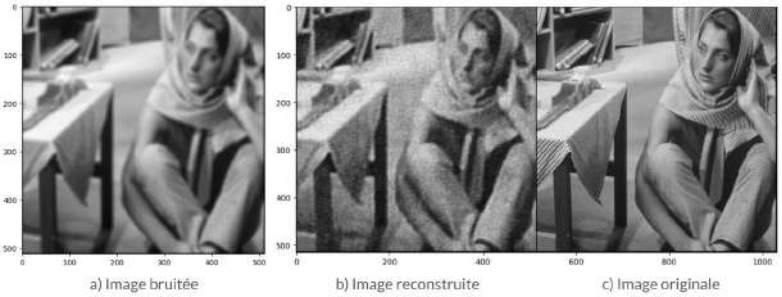


FIGURE 21 – Deconvolution : ResNet Architecture/ 4000 iterations

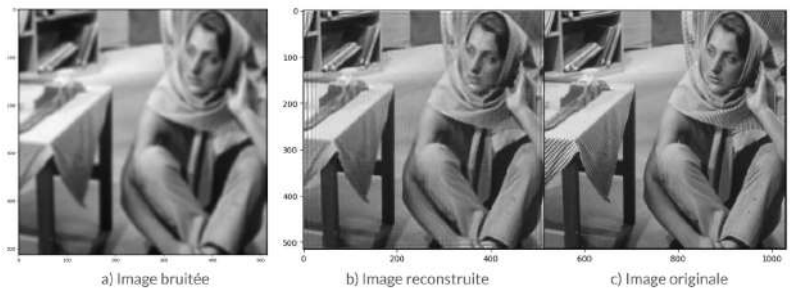


FIGURE 22 – Deconvolution : U-Net Architecture/ 4000 iterations

Comparison

We evaluated the performance of four methods Skip network (autoencoder with skip connections), ResNet, U-Net, and Wiener filter for deconvolution of an image that had been convolved with a smoothing kernel (size 9x9).

The results are summarized in the following table :

Deconvolution	Skip	ResNet	U-Net	Wiener
SSIM	0.6349	0.2009	0.3794	0.60
MSE	0.0049	0.0156	0.0110	0.0060

TABLE 3 – Comparison of Deconvolution Performance Across Architectures (Skip, ResNet, U-Net)



FIGURE 23 – Comparison of the results of the different architectures.

The Skip network achieved the best performance among the deep learning models, with the highest SSIM (0.6349) and the lowest MSE (0.0049). This indicates that the Skip network, with its autoencoder structure and skip connections, was most effective at reconstructing the image while preserving structural details. In comparison, U-Net and ResNet performed less effectively, with U-Net providing a moderate result (SSIM 0.3794, MSE 0.0110), while ResNet showed the lowest performance (SSIM 0.2009, MSE 0.0156), struggling to recover the original image details.



FIGURE 24 – Comparison between the classical method weiner filter and our skip architecture.

The Wiener filter, though a traditional method, also performed well with an SSIM of 0.60 and MSE of 0.0060. While it was competitive, it still could not match the Skip network's ability to preserve image structure. The

Wiener filter demonstrates that classical approaches can still provide useful results, but deep learning methods, particularly those utilizing skip connections, offer superior performance in tasks like image deconvolution.

IV.4 Additive Noise + Convolution Problem

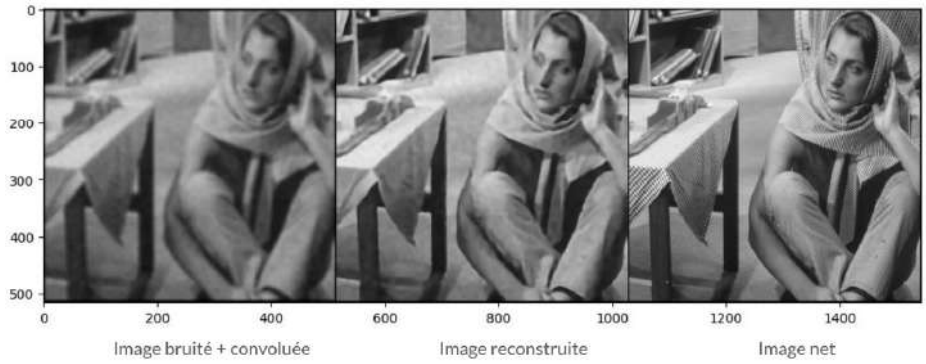


FIGURE 25 – Reconstruction of noisy and convolved images using a Skip Autoencoder architecture with 4000 iterations

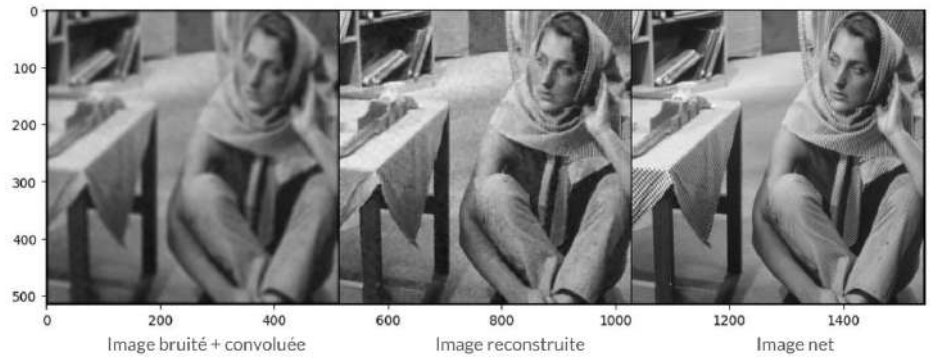


FIGURE 26 – Reconstruction of noisy and convolved images using a Skip Autoencoder architecture with 10000 iterations

We applied the Skip architecture for noise removal, with the image corrupted by additive noise (noise strength $\sigma = 0.15$) and then convolved. Using 4000 iterations, the model achieved an SSIM of 0.608 and an MSE of 0.0047. After increasing the number of iterations to 10000, the performance improved significantly, reaching an SSIM of 0.6245 and an MSE of 0.0042 after 30 minutes of training. However, it is important to stop the training before the model begins to reconstruct the noise, as further training beyond a certain point may lead to overfitting, where the model starts to learn the noise instead of removing it.

V. Conclusion

In this work, we explored the capabilities of the Deep Image Prior approach for various image restoration tasks, including inpainting, deconvolution, and denoising. Through extensive experiments, we demonstrated that even without any external training data, a randomly-initialized convolutional network can effectively reconstruct corrupted images by relying solely on the structure imposed by its architecture. Our results highlight that Deep Image Prior offers a simple yet powerful alternative to traditional methods, especially when the degradation process is unknown or when large datasets are unavailable.

These findings emphasize the significant role of network architecture as an implicit prior. They show that carefully designed architectures can naturally capture important image structures without explicit learning. Overall, Deep Image Prior bridges the gap between handcrafted priors and learned models, offering a new perspective for future research in image restoration and deep learning architectures.

Annexe

NL means

NL (Noise Level) is a parameter that controls the strength or intensity of noise you add to something (like an image, signal, text, or data).

$$\hat{I}(x) = \frac{1}{C(x)} \sum_{y \in \Omega} W(x,y) I(y)$$

- $I(y)$: The value of the pixel at position y.
- Ω : The neighborhood around the pixel x.
- $W(x,y)$: The weight associated with the similarity between patches centered at x and y.
- $C(x)$: Normalization of the weights.

Weiner filter

The Wiener filter is applied in situations where we have a noisy and blurred image and we want to recover the original image. It works by balancing the image's content and the noise.

$$\hat{F}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \frac{5\pi|v|}{s_f(c,v)}} \cdot G(u,v)$$

- $\hat{F}(u,v)$:The Fourier transform of the deconvolved image.
- $H(u,v)$: The transfer function of the system (usually the point spread function in imaging).
- $S_n(u,v)$: The power spectrum of the noise in the noisy image.
- $S_f(u,v)$: The power spectrum of the signal in the noisy image.
- $G(u,v)$: The Fourier transform of the noisy image.

Evaluation metrics

Measure the structural similarity between the two images (SSIM)

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Mean Squared Error (MSE) :

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$